SNB Monetary Policy

Instruments, Variables, and Data

SNB_MoPo project | github.com/ThomasNellen/SNB_MoPo

Data vintage: April 2026

The SNB's Operational Framework

The Swiss National Bank (SNB) implements monetary policy through a **tiered sight-deposit system** combined with a short-term interest-rate target. Since June 2019 the operational target is the **SNB policy rate** (`LZ`), which replaces the earlier three-month CHF LIBOR target range. The policy rate is applied to the bulk of sight deposits held by banks and other eligible institutions at the SNB.

Sight Deposits and the Tiering Mechanism

Banks and certain other financial institutions hold *sight deposits* (overnight balances) at the SNB. The SNB remunerates these deposits at rates that depend on whether a given institution's balance exceeds an individually calculated *exemption threshold*:

- **Up to the threshold** (`ZIGBL`): since September 2022 equal to the policy rate. During the negative-rate era (January 2015 to June 2022) this tranche was exempt from the negative rate and remunerated at 0%.
- **Above the threshold** (`ZIG`): the charge rate applicable to the portion exceeding the threshold. Since June 2022 it equals the policy rate; before that it equalled −0.75%.
- **Threshold factor** (`FREI`): a multiplier applied to each institution's minimum reserve requirement to determine the individual exemption threshold. A higher multiplier means more of a bank's deposits are shielded from the negative rate. It was set at 20 from the introduction of the tiered system in January 2015 until October 2019, then raised repeatedly.

The tiering was introduced in January 2015 alongside the removal of the EUR/CHF floor and the cut of the policy rate (then: LIBOR target midpoint) to −0.75%. Its purpose was to avoid crushing bank profitability by allowing a meaningful portion of reserves to be exempt from negative remuneration. After the SNB exited negative rates in June 2022 the distinction between `ZIGBL` and `ZIG` effectively disappeared (both equal the policy rate), but the legal framework remains in place.

From LIBOR Target to Policy Rate

- **Before June 2019**: the SNB announced a target range for the three-month CHF LIBOR. The effective policy rate is approximated by the rate on sight deposits above the threshold (`ZIG`), which equalled −0.75% throughout this period.
- **From 13 June 2019**: the SNB formally adopted the **SNB policy rate** (`LZ`) as its main operational target. The policy rate replaces the LIBOR midpoint and is directly applied to sight-deposit remuneration.

SARON as the Market Rate

The **Swiss Average Rate Overnight** (SARON) is the overnight reference rate for the CHF money market, computed by SIX as the volume-weighted average of all transactions and binding quotes in the Swiss franc repo market during the trading day. It replaced CHF LIBOR as the standard risk-free reference rate in 2022.

Under normal operating conditions SARON tracks the SNB policy rate closely, typically printing a few basis points *below* the policy rate because banks find it marginally cheaper to transact in the repo market than to hold excess balances at the SNB. Wide deviations between SARON and the policy rate signal stress in the short-term money market.

Policy Decisions

The SNB Governing Board meets quarterly (typically in March, June, September, and December) for a *monetary policy assessment*. Rate decisions are announced on the **Thursday** of the assessment week and take effect on the **following Friday** (the implementation date).

Three types of events are distinguished in the dataset:

Event type	Symbol	Description
`rate_change`	—	Change in ZIG / LZ (the main policy rate)
`rate_intro`	—	Formal introduction of the LZ policy rate (13 June 2019)
`frei_change`	—	Change in the threshold factor (FREI)

Announcement and implementation dates are recorded separately, allowing event-study regressions to distinguish the *announcement effect* from any additional *implementation-day effect* on SARON.

Liquidity Absorption (Quantitative Tightening)

After the SNB lifted rates to positive territory in June 2022, the large stock of excess sight deposits—accumulated through FX interventions—threatened to push SARON below the policy rate. The SNB therefore reactivated and expanded two market-based **liquidity-absorption** instruments: SNB Bills and reverse repos.

SNB Bills (Schatzwechsel)

SNB Bills are short-term, zero-coupon securities *issued by the SNB* and purchased by eligible counterparties (banks, securities dealers, and similar institutions). The purchase transfers CHF from the banking system to the SNB, thereby shrinking excess reserves for the bill's tenor.

Auction mechanics

- **Frequency**: every Thursday.

- **Subscription window**: 11:00–11:30 CET/CEST.
- **Auction type**: Variable Rate, American allotment (VR, A)—each accepted bidder pays their own submitted price.
- **Settlement**: payment four calendar days after the auction ($T+4$); redemption at face value (CHF 1 000 000 per bill) on the maturity date.
- **Maturities**: two tranches per week, always including a **28-day** tranche. The second tranche rotates across 84 d → 168 d → 84 d → 336 d in a roughly four-week cycle, giving four standard tenors in all.
- **Issuance type**: `N` = new ISIN (new series); `A` = tap (additional tranche under an existing ISIN).

History and scale

SNB Bills were first issued in October 2008 as an ad-hoc sterilisation tool during the global financial crisis. They fell dormant during the negative-rate era (2015–2022) and were reactivated in August 2022 alongside the return to positive rates. Since then outstanding volume has grown substantially:

Period	Peak outstanding	Source
2008–2012 (first episode)	CHF ≈25 bn	snbbipo (ES)
2022–2026 (current episode)	CHF ≈99 bn	snbbipo (ES)

Pricing

The marginal yield in SNB Bill auctions is consistently a few basis points *below* the policy rate (typically −5 to −10 bp), reflecting the opportunity cost to banks of committing liquidity for up to twelve months relative to maintaining overnight balances at the SNB. Yields on longer tranches (168 d, 336 d) lie slightly above the 28-day yield, forming a mildly upward-sloping short-end curve.

Reverse Repos (Liquidity-Absorbing Repos)

In a **reverse repo** from the SNB's perspective, the SNB *sells* CHF-denominated securities (typically Confederation bonds or agency paper) to a counterparty under an agreement to repurchase them after the term. The counterparty pays CHF now and receives securities; the SNB receives CHF and holds the repurchase obligation. Net effect: CHF is absorbed from the banking system for the repo term.

This should not be confused with the SNB's *liquidity-providing* repos, in which the SNB *buys* securities from banks (a repo from the bank's perspective).

Auction mechanics

- **Frequency**: every bank working day.
- **Subscription window**: 09:00–09:10 CET/CEST on normal days; delayed to 10:30–10:40 on monetary policy assessment days (approximately quarterly).

- **Standard term**: 7 calendar days (occasionally 14 days around holidays or policy meetings).
- **Settlement**: payment T+2; redemption T+2+term.
- **Auction type**: Variable Rate, American allotment (VR, A).
- **Holiday closures**: no auction on Swiss public holidays; gaps also around Easter, Ascension, and Whitsun.

Data availability

The SNB publishes only a forward-looking auction schedule (updated quarterly, PDF format). *No transaction-level results database exists in the public domain.* Monthly outstanding balances are available from the SNB balance sheet via the `snbbipo` cube (series `VRGSF`).

Variable Definitions

Daily Rate Series (`snbgwdzid` cube)

All rates in per cent per annum; daily frequency.

Code	Name	Available from	Notes
`LZ`	SNB policy rate	13 Jun 2019	Replaced the three-month CHF LIBOR target as the main operational rate. Before this date, ZIG (−0.75%) proxies the effective policy rate.
`ZIG`	Sight-deposit rate above threshold	Jan 2015	Rate applied to deposits exceeding the exemption threshold. Equals LZ since Jun 2022; was −0.75% throughout the negative-rate era.
`ZIGBL`	Sight-deposit rate up to threshold	Sep 2022 (recorded)	Rate applied to deposits within the exemption threshold. Was effectively 0% from Jan 2015 to Jun 2022 (exemption from negative charge); back-filled with 0 in our dataset. Equals LZ since Jun 2022.

Code	Name	Available from	Notes
FREI	Threshold factor	Nov 2019 (recorded)	Multiplier applied to minimum reserve requirements to compute the individual exemption threshold. Was 20 from Jan 2015 to Oct 2019; back-filled with 20 in our dataset.
ZIABP	Discount (basis points)	Sep 2022	Spread between ZIGBL and ZIG in basis points. Currently 0 (rates unified since Jun 2022).
SARON	Swiss Average Rate Overnight	Aug 2009	SIX overnight fixing; volume-weighted average of repo transactions and binding quotes. Main CHF risk-free reference rate since the discontinuation of CHF LIBOR (Dec 2021).
ENG	Special rate (Sondersatz)	May 2004	Rate applied to certain intraday and emergency facilities; not directly relevant to the tiering system.

Policy Decisions Table (`policy_decisions.csv`)

One row per SNB decision event; 24 events from Dec 2014 to Mar 2025.

Column	Type	Description
`announce_date`	date	Thursday of the monetary policy assessment
`implement_date`	date	Friday following the assessment (effective date)
`event_type`	string	`rate_change`, `rate_intro`, or `frei_change`
`ZIG_new`	float	New ZIG rate (%), if changed
`LZ_new`	float	New policy rate LZ (%), if applicable
`FREI_new`	float	New threshold factor, if changed
`label`	string	Short descriptive label for the event

SNB Bills Auction Data (`snb_bills_auctions.csv`)

One row per ISIN per auction; 393 rows covering August 2022 to April 2026.

Column	Type	Description
`auction_date`	date	Date of the auction (Thursday)
`isin`	string	ISIN of the security (CH...)
`denomination_chf`	integer	Face value per bill (CHF 1 000 000)
`issuance_type`	string	`N` = new ISIN; `A` = tap
`term_days`	integer	Exact calendar days from payment to redemption
`payment_date`	date	Settlement date (T+4)
`redemption_date`	date	Maturity date
`auction_type`	string	Always "VR, A" (variable rate, American)
`marginal_price_pct`	float	Marginal accepted price as % of face value
`yield_pct`	float	Yield at the marginal price, per cent p.a.
`bids_chf_mn`	float	Total bids submitted (CHF mn); published end of month
`allotment_chf_mn`	float	Total allotment to market (CHF mn)
`allot_issuer_chf_mn`	float	SNB's own retention (CHF mn)
`outstanding_chf_mn`	float	Outstanding volume for this ISIN after auction (CHF mn)
`term_bucket`	string	Rounded tenor: `7d`, `28d`, `84d`, `168d`, `336d`

Note on publication timing: the SNB publishes the marginal price on the first bank working day after the auction at 18:00. Bids, allotment, and outstanding volumes are published on the last bank working day of the *following* month; rows for the most recent auctions therefore show `NaN` for those columns until the publication date.

Weekly Bills Aggregation (`snb_bills_weekly.csv`)

One row per auction date; allotment summed by term bucket.

Column	Type	Description
`auction_date`	date	Auction Thursday
`7d`	float	Allotment in the 7-day tranche (CHF mn)
`28d`	float	Allotment in the 28-day tranche (CHF mn)
`84d`	float	Allotment in the 84-day tranche (CHF mn)
`168d`	float	Allotment in the 168-day tranche (CHF mn)
`336d`	float	Allotment in the 336-day tranche (CHF mn)
`total_allotment_chf_mn`	float	Sum across all tranches (CHF mn)

Monthly Balance Sheet (`snb_balance_sheet_monthly.csv`)

Monthly end-of-period balance sheet positions from the `snbbipo` cube, CHF millions.

Column	SNB code	From	Description
`snb_bills_outstanding_chf_mn`	`ES`	Oct 2008	Outstanding SNB Bills (liability side). Sum of all live ISINs.
`reverse_repo_liabilities_chf_mn`	`VRGSF`	Apr 1998	Reverse repo liabilities in CHF: SNB's obligation to repurchase securities sold under repo. Measures cumulative liquidity absorbed through reverse repos.
`repo_claims_chf_mn`	`FRGSF`	Apr 1998	Repo claims in CHF: SNB's claims from liquidity-*providing* repos (asset side). Not a QT instrument.
`sight_deposits_chf_mn`	`N`	Dec 1996	Sight deposits of domestic banks (liability side). Main stock of excess reserves in the banking system.
`total_assets_chf_mn`	`TO`	Dec 1996	Total assets of the SNB balance sheet.
`banknotes_chf_mn`	`UA`	Dec 1996	Banknotes in circulation.
`total_liq_absorption_chf_mn`	`ES` + `VRGSF`	Oct 2008	`ES` + `VRGSF`: total liquidity absorbed through bills and reverse repos combined.

Data Sources and Reproducibility

Source	URL / reference
SNB Data Portal	https://data.snb.ch
Daily rates cube	https://data.snb.ch/api/cube/snbgwdzid/data/csv/en
Balance sheet cube	https://data.snb.ch/api/cube/snbbipo/data/csv/en
SNB Bills results	https://www.snb.ch/public/bills/snbBillsResults.pdf
Repo auction schedule	https://www.snb.ch/public/repos/repoAuctionSchedule.pdf
Bills calendar	https://www.snb.ch/public/bills/snbBillsCalendar.pdf
Policy decisions	Compiled manually from SNB press releases

All data are downloaded and processed by two scripts in the `code/` directory:

- `snb_mopo_analysis.py`: daily rates, policy decisions, and the tiered-system figure.

- `snb_qt_analysis.py`: SNB Bills auction data and monthly balance sheet positions.

Running either script from the project root reproduces the processed datasets in `input/processed/` from scratch. Raw downloads are cached in `input/raw/` (gitignored) and only re-downloaded if absent.

Figure: Tiered Sight-Deposit Remuneration and SARON

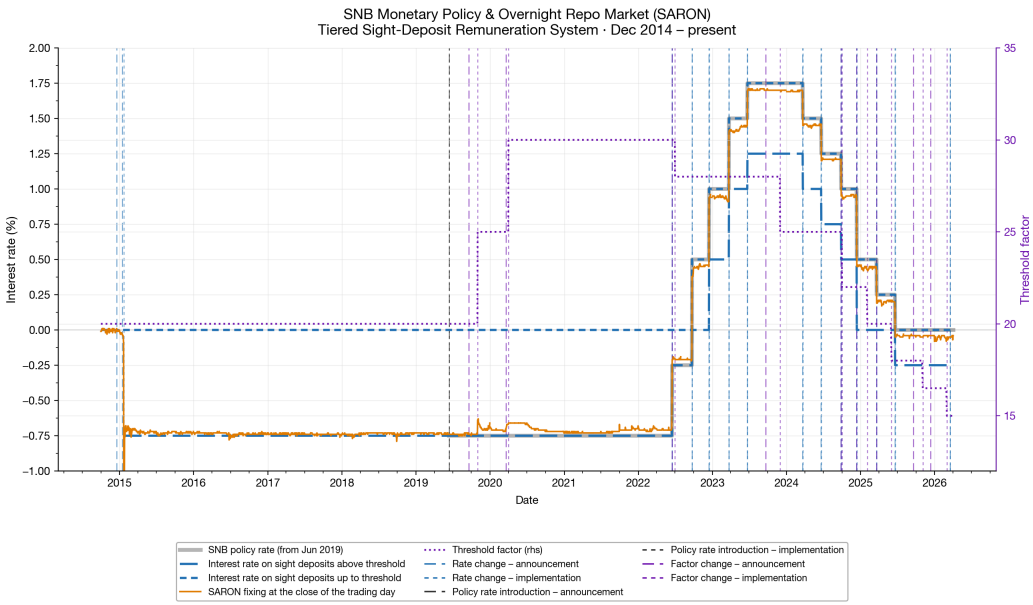

Figure 1: SNB policy rates, sight-deposit remuneration tiers, threshold factor (right axis), and SARON since December 2014. Vertical dashed lines mark SNB announcement dates (long dashes) and implementation dates (short dashes), coloured by the type of change: steel blue for rate changes, charcoal for the policy-rate introduction, and purple for threshold-factor changes. Source: SNB Data Portal (`snbgwdzid`); own computation.